What Do People Optimize in Diagram Layout?

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Abstract

Fitting a layout energy to human drawings by inverse optimisation assumes the drawings sit at a minimum of some weighted sum of the standard criteria. We test that assumption one criterion at a time on 853 hand-drawn diagrams of 15 to 40 boxes (WikiPathways, Reactome, BPMN): move each box 2% of the drawing width in sixteen directions and ask whether any move lowers the criterion, `neato`, `dot` and ELK layouts of the same graphs as controls. Overlap holds every box at zero, by hand and in every tool that removes it. Uniform edge length and stress hold almost no hand-placed box, 0.3 to 1.1% per corpus and a median diagram of 0.00, and not for want of anything to gain: every diagram has improving moves, the best of them worth 1.2% of the length term and 0.6% of stress. `neato`'s stress layouts are held at 1.00 under the same test, so it has power; a weight of 0.001 on either term drops the held fraction below 0.015. The criterion that holds a hand layout and that the energy omits is alignment into rows and columns: 0.52 and 0.21 of boxes on the two biological corpora and 0.87 over the full BPMN population, against 0.06 for `neato`, 1.00 for `dot`, and 0.07 both for a layout snapped to the same grid with no author and for a uniform random layout of the same graphs. Whether the person's hand or the editor's alignment guide supplies those rows, the coordinates cannot say, and for fitting an energy it does not matter: the term is held and absent either way. A hand layout is a layered layout with partial alignment. The pattern is not the diagrams': 136 drawings of abstract graphs from HOLA's formative study hold alignment at 0.50 and the distance terms at 0.00. Coordinates and judgement disagree: judges have been found to track stress, and the coordinates people draw are held by no distance term we tested.

1 Introduction

A layout program places the boxes of a diagram by minimising an *energy*, a weighted sum of quantities the field has agreed are bad: edges that cross, boxes that overlap, edges of very unequal length, boxes that sit on edges they do not belong to [3, 2, 4, 5]. The weights are asserted by the tool's author, and since 1994 the proposal has been to learn them instead, from layouts people made or preferred [15, 26, 14, 19, 28]. What such a method needs of its examples varies, and the difference matters here. Learning from preferences or selections [15, 26, 14, 28] needs only that an accepted layout rank well under the learned energy. Inverse optimisation, which fits an objective to solutions taken as optimal [22, 23, 24], needs the stronger property: that the coordinates themselves sit at a minimum. The stronger property has never been tested against human coordinates, no published layout method yet commits to it, and 4,890 hand-made drawings were just collected into one corpus [12], so the first coordinate fit is one download away. It is the property that separates an energy describing where a person put a box from an energy describing how they ranked the result. Where it fails, the fitted weights are the criteria's mutual

*Independent researcher. Code, parsed corpora and every table's script: `https://github.com/Anode1/cjitter`, directory `example/diagrams`.

trade-offs rather than the person's, and the term the person did optimise stays unlearned because it was never offered.

The assumption has a direct test, one criterion at a time. Take a diagram a person drew. Move one box a small distance in each of several directions and ask whether any of the moves lowers the criterion. If none does, the box is *held*: the person has put it where the criterion wants it, or where the criterion has stopped caring. Do this for every box and the fraction held says how much of the layout is a local minimum of that criterion; the criterion's value says how well it is met. A criterion that holds every box at value zero is one the person satisfied; one that holds no box is one the person did not consider, or traded away.

What the test finds puts coordinates and judgement on opposite sides. Mooney et al. [11] report that people prefer low-stress drawings and locate shortest paths better as stress falls, on graphs to 50 nodes; Dwyer et al. [17] had seen the preference half earlier, over twelve layouts and without a test. The coordinates people drew here are held by no distance term at all, stress included, in any of three corpora, while `neato`'s stress layouts are held at 1.00 under the same test. People prefer what they do not draw, and a criterion can be wanted in the result without being visible in the coordinates.

We ran the test on 853 diagrams of 15 to 40 boxes from three unrelated communities (Section 4), and on layouts of the same graphs by four tools, which say what a layout that does minimise something scores. Section 6 carries the numbers; the findings are these.

1. Overlap holds every box of the median hand diagram at value zero in every corpus, as it does in every tool that removes overlaps; the term is nonzero in 4 to 8% of hand diagrams. People satisfy it exactly.
2. Length and stress hold a hand-placed box almost nowhere, median diagram 0.00 in all three corpora and 25 to 78 boxes of roughly 7,000 each, while `neato`'s stress layouts are held at 1.00. No reweighting repairs it: a weight of 0.001 on either term drops the held fraction of the sum below 0.015.
3. Alignment into rows and columns holds 0.52, 0.21 and 0.87 against `neato`'s 0.06 and 0.07 for a grid-snapped random layout. It is the criterion the base energy omits and the person supplies, and BPMN's lanes cannot impose it.
4. Reading direction does not close the gap. Flow holds 0.59 to 0.85 by hand where `dot` and ELK hold 1.00: the person tolerates a backward edge the layered tools remove.
5. Crossings and node–edge separation hold every box because they are flat at this radius, not because they are met. Only 3 to 30% of diagrams offer any improving move.
6. On everything else a hand layout reads as a layered layout: held under overlap, free under distance, partly held under orthogonality.

The rest of the paper is the foundation behind the test (Section 2), the instrument (Section 3), the corpora, the method, the measurements, the alternatives not yet excluded (Section 7), and what is already known (Section 8). A tool builder reading only this far has the changes: add alignment to the energy and print the fraction of boxes it holds when started from a layout a person accepted; edge length keeps its weight, for the reason the conclusion gives.

2 Foundations

The energy. A layout of a graph with N nodes is a point $x \in \mathbb{R}^{2N}$, two coordinates per node, the boxes' sizes held fixed. An aesthetic energy is

$$E_w(x) = \sum_{k=1}^{K} w_k \, T_k(x), \qquad w_k \geq 0, \tag{1}$$

where each term T_k is a quantity to be made small. The tool's author asserts the w_k; the field of multicriteria layout is the choice of the T_k and the w_k [2, 4].

Local minimum. A point x is a local minimum of E if $E(y) \geq E(x)$ for every y within some distance r of x; for differentiable E this implies $\nabla E(x) = 0$. Two terms here are not differentiable, crossings being a count and overlap having corners where boxes touch, so the test uses no gradient. Fix a radius d and a set V of unit directions in the plane (sixteen, equally spaced). For node i and direction v let $x + d\,v^{(i)}$ be the layout with node i moved by d along v and every other node fixed. Node i is *held* at radius d if

$$E(x + d\,v^{(i)}) \geq E(x) \quad \text{for every } v \in V, \tag{2}$$

and the estimand is

$$q_d(x) = \frac{1}{N}\,\#\{\, i : \text{node } i \text{ is held at radius } d\,\}. \tag{3}$$

$q_d = 1$ says the layout is a local minimum of E over single-node moves of length d; for smooth E and small d it says $\nabla E(x) = 0$ to within the resolution of V. Being a comparison of two energies, (2) is defined for every term and, evaluated per term, separates the criteria a person satisfied from those they did not. A capped descent, this study's first estimand, answers the same question for the sum only; it is retained as a secondary.

The inverse problem. Given layouts $x^{(1)}, \ldots, x^{(G)}$ that people accepted, inverse optimisation asks for the weights w under which they are nearest to stationary [23, 24]. With the directional differences $D_{iv}T_k = T_k(x + d\,v^{(i)}) - T_k(x)$ in hand, node i is held under w exactly when $\sum_k w_k D_{iv} T_k \geq 0$ for all v, a set of linear inequalities in w. The residual is the smallest total violation over w on the simplex, a linear program, and it is zero only if some weighting of the terms makes the layout a local minimum. A term with $D_{iv}T_k < 0$ for some v at almost every node (uniform edge length, at any layout whose edges are not all the same length) cannot take positive weight without violating the inequalities at those nodes, which is the zero weight the random-search fit found.

Step terms. A count (crossings) or a step function of position (gridiness, Section 5) is flat almost everywhere and holds a node wherever no move of length d changes it. There q_d measures how much of the layout the criterion has stopped acting on, not how well it is met, so the term's value is reported beside q_d.

The reference length. The length and stress terms compare distances with a reference L. Fixed at the layout's median edge length, L is in general not the scale at which the term is least for that layout, so at a true minimum every node still has a scaling direction that lowers the term and the converged reference is not held (neato's stress layouts: $q_{0.02} = 0.07$ to 0.12). L is therefore the value at which the term is least for the layout as loaded, $\sum \ell^2 / \sum \ell$ over edges for length and the same over $\|x_i - x_j\|/\delta_{ij}$ for stress, fixed before any move; neato's stress layouts are then held at 1.00 in every corpus. The median and $1/\sqrt{N}$ are sensitivities.

This is not a new device and the paper does not claim it as one. Minimising a stress term over a scale factor is Smelser, Miller and Kobourov's scale-normalised stress [9], who give the minimiser in closed form and read it as a correlation coefficient, and Ahmed et al. [10] derive the same for the δ_{ij}-weighted graph stress used here. Their result is also why the fixed-L reading above is not a curiosity: under scale-sensitive stress they find a random layout ranked ahead of both `neato` and SFDP on essentially every graph of two 1,000-graph benchmarks. A control that reads 0.07 at a fixed L is that pathology, and fitting L is the known repair.

The null and the controls. A test that can only succeed is not a test. The *stationarity reference*: q_d on a layout a descent has converged on must be near one, or the test lacks power for that energy; neato's layouts are the reference for stress (1.00 in every corpus) and the uncapped climber's end point for the sums (median 1.00 for every energy and corpus, Section 6). The *tool controls*: q_d on layouts of the same graph with the same box sizes by `neato` (stress, boxes left where they overlap), `prism` (neato followed by its overlap removal) and `dot` (layered), which say what a layout that does minimise something scores on each term. The *matched-budget control*: uniform random placement at the same number of energy evaluations, which the hand layout must beat, or "held under overlap" would be a property of sparse graphs and not of the person.

3 The instrument

The descents, the control and the paired statistics come from `cjitter`, a C library of four stochastic searches over a box of real variables: hill climbing with restarts, simulated annealing, a genetic algorithm, and uniform sampling as the control. Every run is byte-reproducible from its seed on every platform, every method constant is a visible field, and `cjitter_compare` runs all four at the same budget on the same seeds and reports an exact paired sign test, Holm-corrected, against the control. "Better than uniform sampling at equal cost" is the only claim it lets a search make.

Its `block` operator, one object moved a random step and the step kept when the summed energy fell, is the operator of a label placer the author deployed in 2001; the directional test of equation (2) is that proposal made exhaustive, every direction tried, none accepted, the answer read off.

4 The corpora

Three collections of diagrams whose node coordinates a person chose, from three communities with different authoring tools, all redistributable.

corpus	licence	files	parsed	15–40 nodes	used	median m/n
WikiPathways GPML, *Homo sapiens*	CC0	1,016	571	305	305	1.06
Reactome SBGN-ML	CC0	1,382	710	248	248	1.04
BPMN Academic Initiative	academic	29,810	21,784	6,723	300	1.11

The layouts are working documents: Reactome diagrams are laid out by curators [32], WikiPathways pathways are drawn in PathVisio [33], and the BPMN models are students' work in Signavio [34, 35]. Nobody was asked to draw nicely. The three tools share no layout code. Two WikiPathways files (WP5037, WP3391) carry coordinates converted from Reactome [36] and are dropped.

The BPMN archive is mixed: of the 21,784 models the parser reads, 13,435 have every edge a BPMN flow or association and the rest are EPC, Petri net and UML models, whose flow elements carry stencils outside the connector set and so become degree-2 nodes under the node rules. Only the 13,435 enter, 6,723 of them in the band, and the first 300 by model id are used; the two biological corpora are used whole. An earlier version of this study took 147 models without the filter, two fifths of them read as bipartite graphs.

A diagram enters as its largest connected component (194 of the 248 Reactome graphs and 120 of the 300 BPMN graphs in the band are the component of a larger diagram; no WikiPathways pathway in the band is); container glyphs (SBGN compartments, BPMN pools and lanes) are not nodes; SBGN process glyphs are nodes, so the Reactome graphs are bipartite and their m/n is the process graph's; each layout is rescaled so that its bounding box fits the unit square, box sizes scaled with it. The directional test runs on 15 to 40 nodes; the climber pilot used 10 to

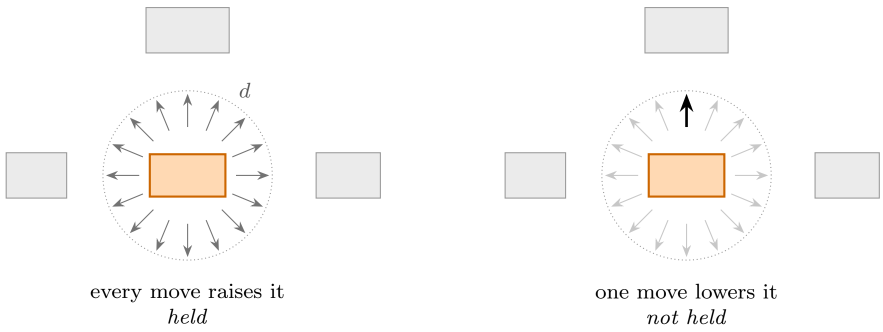


Figure 1: The test, one box and one criterion. The box is moved a distance d, two per cent of the drawing's width, in each of sixteen exact directions, and the criterion is recomputed. A box no move improves is held: the person put it where the criterion wants it, or where the criterion has stopped caring. The distinction between those two costs the paper a term-by-term reading, since a criterion flat at this radius holds every box without being satisfied by any of them.

400. Diagrams are not deduplicated: the WikiPathways set is *Homo sapiens* only, so it holds no ortholog copies, and no check was made for BPMN models submitted twice. The 41 to 100 node band was not run.

How the coordinates were made matters for the alignment criterion. Of the raw coordinates, 17% are integers in WikiPathways, 67% in Reactome and 63% in BPMN (44% multiples of five); the fraction of boxes sharing an exact x with another box is 0.37, 0.06 and 0.42, an exact y 0.29, 0.12 and 0.76, and either 0.54, 0.17 and 0.88, which is the figure the alignment row of Section 6 should be read against. In WikiPathways exact alignment is twice as common as integer coordinates, so it is not a rounding grid's act; whether it is the person's hand or the editor's snap-to-object guide, which PathVisio draws on every drag, the coordinates cannot say. BPMN authors work on a grid inside lanes, which is where the y figure of 0.76 comes from, and their editors guide every drag toward the neighbours; Reactome's curation tool rounds and its authors align little. The hold orders exactly as the editors' affordances do, which Section 7 takes up.

5 The method

Terms. With n nodes, m edges, box i of width w_i and height h_i, L the reference length of Section 2, fitted to the layout under test:

term	definition	zero when
T_C crossings	proper crossings between segments of two routes with no shared endpoint, $/m$	planar
T_O overlap	overlap area of boxes / total box area	no overlap
T_L length uniformity	mean over edges of $((\ell - L)/L)^2$	all edges of length L
T_R orthogonality	mean over edges of the length-weighted mean over its segments of $\min(\lvert\Delta x\rvert, \lvert\Delta y\rvert)/(\lvert\Delta x\rvert + \lvert\Delta y\rvert)$	routes axis-aligned
T_A alignment	$1-$ fraction of nodes in an alignment of ≥ 3 (tol. 0.005)	every node aligned
T_N node–edge	squared shortfall of node to foreign route distance below $(w+h)/4$	none within
T_F flow	mean over directed edges of the backward component along the reading direction u, over length	no edge runs back

The base energy is $T_C + T_O + T_L$ with weights $(1, 1, 1)$, the form of the simulated-annealing layout of Davidson and Harel [2] and of the multicriteria tradition since [29]. The field's other default is stress [30, 31], $\sum_{i<j}(\lVert x_i - x_j\rVert - L\,\delta_{ij})^2/\delta_{ij}^2$ with δ_{ij} the graph distance, which force-directed tools and `neato` minimise and which Dwyer et al. [17] found human choices track; it is the second base energy. T_R, T_A and T_N are the candidates for what the base energy misses.

The alignment definition above is HOLA's gridiness [13], the published form, a step function of position. Three continuous forms are the sensitivities: A1, per node the distance to the nearest other node along whichever axis is nearer, which has a corner at exact alignment; A2, rows and columns priced separately; A3, a Gaussian kernel of width $s = 0.02$, smooth. The corner is what holds a box: A1 holds 0.52 of WikiPathways boxes and A3 holds 0.20.

Routes. An edge is the polyline the file stores: the attachment point on the source box, the interior waypoints, the attachment point on the target box; where the file stores nothing it is the chord between the centres. Crossings, orthogonality and node–edge separation are evaluated on the route; length and stress on the centres. When a node moves its attachment points move with it and the waypoints stay. The attachment point matters: of the 2,584 BPMN routes that carry an interior waypoint, 0.27 are axis-aligned at every segment when the route is closed at the box centres and 0.88 when it is closed at the stored attachments. What each format stores, as a fraction of the edges in the band:

corpus	edges	with a waypoint	by what the file stores
WikiPathways	8,253	0.291	chord 0.587, anchor 0.225, elbow 0.154, curved 0.026, segmented 0.008
Reactome	6,827	0.069	chord 0.931, stored 0.069
BPMN	7,910	0.327	chord 0.673, stored 0.327

GPML's Elbow and Curved connectors store their endpoints only and PathVisio computes the bends at draw time, so those routes read as chords; a quarter of WikiPathways edges end on an anchor of another interaction, where the person drew no segment to the node and the last segment is manufactured. Only the BPMN routes are orthogonal: of the WikiPathways and Reactome routes with a waypoint, 0.009 and 0.007 are axis-aligned at every segment, the biological formats drawing diagonal segments. The chords are the sensitivity.

Reading direction. T_F needs a direction u. It is one of the four axis directions, the one under which T_F is least for the layout as loaded, fixed before any move, the same rule as for L. Undirected edge kinds (GPML graphical lines, BPMN undirected associations) do not enter it.

Directional test. Radius $d \in \{0.005, 0.01, 0.02, 0.05\}$ of the drawing width, sixteen directions, every node, each term alone and the sums of the tables. Primary cell $d = 0.02$. The directions are exact to double precision (a series evaluated in the first quadrant and rotated), and a move that lowers the energy by less than 10^{-12} of its value is a tie; an earlier build with directions off by up to 4×10^{-6} counted the shift that error made as an improvement and read alignment lower by up to 0.04. Deterministic: no seed, no budget. Every median carries a 95% bootstrap interval over diagrams (1000 resamples).

Tool controls. The same graphs with the same box sizes and edge directions laid out by `neato` (`-Gstart=1`, so the file reproduces), by `neato` with `-Goverlap=false` (prism), by `dot`, and by ELK layered 0.12.0 (`elkjs`, direction RIGHT, every other option at its default), the engine the BPMN and KIELER editors run; a check in the binary confirms each control is the same graphs.

Descent, secondary. Through the library: hill climbing from the hand layout, one node per proposal, first step $d/2$ per coordinate, the step halved after 200 rejected proposals, each node kept within a disc of radius d about its start by the repair, 4000 energy evaluations, 15 seeds per instance; and uniform sampling in the same cap at the same budget on the same seeds, the matched-budget control. Reported per instance: the fraction of the cap used (mean over nodes of the distance moved over d), the fraction of energy removed, for the climber and for the control, and the number of seeds on which the climber ends lower than the control with its exact sign test. The climber pilot used a step of 0.1, five times the cap, so that nearly every proposal landed on

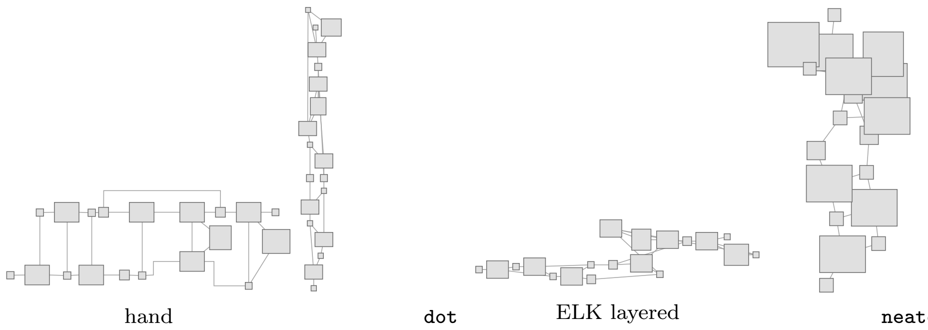


Figure 2: The same graph, the same box sizes, four layouts. The hand drawing and the two layered tools share the profile the test reads: held under overlap, free under length and stress. `neato` minimises stress and is the control that gives the test its power, held at 1.00 under stress where every hand layout reads 0.00. The tool layouts carry no waypoints, so their edges are drawn as chords.

the cap circle; the base-energy result did not depend on it (98.5 to 95.9% of the cap used as the step fell from 0.1 to 0.005) and the alignment result did (47 to 18%), which is why the directional test is primary. The converged reference is the same climber uncapped, 60,000 evaluations, with q_d measured where it stops.

Fitting. Weights on the simplex, the linear program of Section 2 over every node and direction of a corpus (about 120,000 inequalities, HiGHS), the residual its value per node; because the residual depends on the terms' scales and q does not, q under the fitted weights is reported beside it, on the fitting diagrams and held out under 5×2 cross-validation (five seeded halvings, each half fitted and scored on the other). Inverse optimisation is degenerate whenever a term is satisfied exactly: overlap holds every node, so any weight vector with all its weight on overlap has residual zero and the program returns it. The informative output is the sweep, q as the weight on one term grows from 0 with the rest on crossings and overlap, which is the test of whether any positive weight on that term can be carried.

Statistics. The unit of inference is the diagram. Per corpus, the paired comparison of q_d between the hand layout and a tool control is the Wilcoxon signed-rank test, zeros dropped, ties by midranks, exact on the observed ranks up to 50 non-zero pairs and the normal approximation with continuity and tie corrections above, with the exact sign test beside it and the Hodges–Lehmann shift with its distribution-free 95% interval; the family is the terms compared against one control within one corpus, Holm-corrected. The script carries 630 self-checks against enumeration and against R. Equivalence claims ("no reweighting repairs it") are read off the sweep, never off a failure to reject.

6 Measurements

Every number in this section is from one battery on the corpora of Section 4, run 2026-08-23 on commit `1987633` (`example/diagrams/results/run_all.sh`; versions, seeds and corpus hashes in `results/manifest.txt`). The directional test is one deterministic pass over the whole band on the stored routes with the fitted L.

Each criterion alone. $q_{0.02}$ on the hand layout, with its 95% bootstrap interval over diagrams, and on the three tool layouts of the same graphs; in parentheses the median value of the term at the layout, 0 when satisfied.

corpus	energy	hand	`neato`	`prism`	`dot`	`elk`
WikiPathways	crossings alone	1.00 [1.00, 1.00] (0.000)	1.00 (0.000)	1.00 (0.000)	1.00 (0.029)	1.00 (0.036)
	overlap alone	1.00 [1.00, 1.00] (0.000)	0.60 (0.030)	1.00 (0.000)	1.00 (0.000)	1.00 (0.000)
	length alone	0.00 [0.00, 0.00] (0.219)	0.24 (0.011)	0.19 (0.014)	0.00 (0.241)	0.00 (0.127)
	stress alone	0.00 [0.00, 0.00] (0.217)	1.00 (0.042)	0.94 (0.044)	0.00 (0.201)	0.00 (0.241)
	orthogonality alone	0.19 [0.18, 0.21] (0.157)	0.00 (0.280)	0.00 (0.281)	0.19 (0.244)	0.19 (0.158)
	alignment A1 alone	0.52 [0.48, 0.54] (0.003)	0.06 (0.011)	0.07 (0.010)	1.00 (0.000)	0.57 (0.001)
	alignment A3 alone	0.20 [0.18, 0.23] (0.284)	0.03 (0.446)	0.03 (0.451)	0.45 (0.204)	0.35 (0.176)
	gridiness alone	0.87 [0.84, 0.89] (0.424)	0.78 (1.000)	0.79 (1.000)	1.00 (0.125)	0.76 (0.200)
	node-edge alone	0.88 [0.86, 0.90] (0.004)	0.89 (0.000)	1.00 (0.000)	0.78 (0.011)	0.74 (0.016)
	flow alone	0.59 [0.56, 0.62] (0.131)	0.50 (0.211)	0.50 (0.211)	1.00 (0.000)	1.00 (0.000)
Reactome	crossings alone	1.00 [1.00, 1.00] (0.000)	1.00 (0.000)	1.00 (0.000)	1.00 (0.056)	1.00 (0.066)
	overlap alone	1.00 [1.00, 1.00] (0.000)	0.76 (0.022)	1.00 (0.000)	1.00 (0.000)	1.00 (0.000)
	length alone	0.00 [0.00, 0.00] (0.234)	0.35 (0.005)	0.26 (0.009)	0.00 (0.232)	0.00 (0.205)
	stress alone	0.00 [0.00, 0.00] (0.212)	1.00 (0.034)	0.85 (0.042)	0.00 (0.177)	0.00 (0.197)
	orthogonality alone	0.17 [0.15, 0.18] (0.193)	0.00 (0.277)	0.00 (0.277)	0.15 (0.256)	0.12 (0.203)
	alignment A1 alone	0.21 [0.19, 0.24] (0.004)	0.06 (0.009)	0.06 (0.010)	1.00 (0.000)	0.42 (0.002)
	alignment A3 alone	0.10 [0.08, 0.12] (0.300)	0.03 (0.430)	0.03 (0.428)	0.39 (0.206)	0.31 (0.213)
	gridiness alone	0.65 [0.62, 0.68] (0.640)	0.77 (1.000)	0.75 (1.000)	0.95 (0.176)	0.74 (0.306)
	node-edge alone	1.00 [0.94, 1.00] (0.000)	0.93 (0.000)	1.00 (0.000)	0.80 (0.007)	0.75 (0.021)
	flow alone	0.62 [0.60, 0.67] (0.096)	0.45 (0.217)	0.44 (0.222)	1.00 (0.000)	1.00 (0.000)
BPMN	crossings alone	1.00 [1.00, 1.00] (0.000)	1.00 (0.000)	1.00 (0.000)	1.00 (0.000)	1.00 (0.000)
	overlap alone	1.00 [1.00, 1.00] (0.000)	0.41 (0.073)	1.00 (0.000)	1.00 (0.000)	1.00 (0.000)
	length alone	0.00 [0.00, 0.00] (0.291)	0.42 (0.006)	0.27 (0.014)	0.00 (0.250)	0.00 (0.273)
	stress alone	0.00 [0.00, 0.00] (0.203)	1.00 (0.017)	0.90 (0.024)	0.00 (0.169)	0.00 (0.190)
	orthogonality alone	0.73 [0.70, 0.75] (0.011)	0.00 (0.280)	0.00 (0.278)	0.26 (0.167)	0.25 (0.139)
	alignment A1 alone	0.91 [0.87, 0.97] (0.000)	0.06 (0.011)	0.06 (0.011)	0.89 (0.000)	0.62 (0.001)
	alignment A3 alone	0.31 [0.29, 0.33] (0.207)	0.00 (0.453)	0.00 (0.452)	0.19 (0.168)	0.20 (0.155)
	gridiness alone	0.95 [0.94, 0.97] (0.176)	0.78 (1.000)	0.78 (1.000)	0.79 (0.219)	0.69 (0.250)
	node-edge alone	1.00 [1.00, 1.00] (0.000)	1.00 (0.000)	1.00 (0.000)	0.82 (0.003)	0.76 (0.016)
	flow alone	0.85 [0.82, 0.87] (0.045)	0.62 (0.121)	0.60 (0.123)	1.00 (0.000)	1.00 (0.000)

The sums. The same test under the summed energies, equal weights.

corpus	energy	hand	`neato`	`prism`	`dot`	`elk`
WikiPathways	C+O	1.00 [1.00, 1.00]	0.60	1.00	1.00	1.00
	C+O+L	0.00 [0.00, 0.00]	0.17	0.19	0.00	0.00
	C+O+S	0.00 [0.00, 0.00]	0.60	0.94	0.00	0.00
	C+O+R	0.21 [0.19, 0.23]	0.00	0.00	0.19	0.22
	C+O+A1	0.51 [0.47, 0.53]	0.04	0.07	1.00	0.57
	C+O+A3	0.24 [0.22, 0.27]	0.03	0.03	0.50	0.41
	C+O+grid	0.84 [0.81, 0.87]	0.45	0.79	0.96	0.74
	C+O+N	0.86 [0.83, 0.88]	0.55	1.00	0.78	0.74
	C+O+F	0.56 [0.53, 0.59]	0.30	0.50	1.00	1.00
Reactome	C+O	1.00 [1.00, 1.00]	0.76	1.00	1.00	1.00
	C+O+L	0.00 [0.00, 0.00]	0.33	0.27	0.00	0.00
	C+O+S	0.00 [0.00, 0.00]	0.76	0.89	0.00	0.00
	C+O+R	0.17 [0.16, 0.19]	0.00	0.00	0.14	0.13
	C+O+A1	0.22 [0.19, 0.25]	0.05	0.07	1.00	0.43
	C+O+A3	0.11 [0.09, 0.12]	0.03	0.03	0.42	0.33
	C+O+grid	0.64 [0.61, 0.67]	0.52	0.75	0.94	0.73
	C+O+N	0.94 [0.91, 0.95]	0.71	1.00	0.80	0.75
	C+O+F	0.61 [0.58, 0.65]	0.33	0.44	1.00	1.00
BPMN	C+O	1.00 [1.00, 1.00]	0.41	1.00	1.00	1.00
	C+O+L	0.00 [0.00, 0.00]	0.24	0.29	0.00	0.00
	C+O+S	0.00 [0.00, 0.00]	0.43	0.92	0.00	0.00
	C+O+R	0.73 [0.71, 0.75]	0.00	0.00	0.28	0.40
	C+O+A1	0.89 [0.86, 0.95]	0.04	0.09	0.88	0.68
	C+O+A3	0.33 [0.30, 0.37]	0.03	0.03	0.33	0.45
	C+O+grid	0.95 [0.93, 0.96]	0.28	0.78	0.75	0.67
	C+O+N	1.00 [1.00, 1.00]	0.40	1.00	0.82	0.76
	C+O+F	0.84 [0.81, 0.87]	0.26	0.61	1.00	1.00

Interpretation of the sums. Read down a hand column. Overlap holds every box at value zero; so it does in every tool layout that removes overlaps, and `neato` without removal is the only layout it does not hold, so crossings and overlap together separate a hand layout from that configuration and from nothing else. Length and stress hold almost no hand-placed box, 0.3 to 1.1 per cent per corpus, and `neato`'s stress layouts are held under stress at 1.00, so the zero is the layout's and not the test's; `dot`, which is layered and prices no distance, scores 0.00 on both as the hand layouts do. At the hand layout the three base terms share the energy as length 0.99, crossings 0.01, overlap 0.00, and $C + O + L$ inherits the length term's zero. Alignment A1 is where hand and `neato` part furthest, and a layer being a row is why `dot` and ELK score high on it too; a layout snapped to the same grid with no author holds 0.065, 0.075 and 0.067 (Section 7). What q counts under T_A is the fraction of boxes within $d/2$ of sharing a coordinate, not the fraction exactly on one, the term having its corner there, which is why it tracks the exact-tie fractions of Section 4 to within a few points.

What q counts, term by term. A held fraction is not commensurable across terms of different shape, and reading down a column compares three different things. Beside each q the table gives *dec*, the mean over boxes of the best decrease a single move finds as a fraction of the term's value, and *active*, the share of diagrams in which any box has an improving move at all. Where *active* is small the term is flat at this radius: it holds every box without being satisfied by any of them, and its 1.00 carries no information about the person.

	WikiPathways			Reactome			BPMN		
term	q	dec	active	q	dec	active	q	dec	active
crossings	1.00	0.000	30%	1.00	0.000	20%	1.00	0.000	3%
overlap	1.00	0.000	18%	1.00	0.000	4%	1.00	0.000	7%
node–edge	0.88	0.047	64%	1.00	0.000	49%	1.00	0.000	15%
length	0.00	0.013	100%	0.00	0.013	100%	0.00	0.012	100%
stress	0.00	0.005	100%	0.00	0.006	100%	0.00	0.007	100%
orthogonality	0.19	0.012	100%	0.17	0.027	100%	0.72	0.008	99%
alignment A1	0.52	0.066	92%	0.21	0.069	99%	0.91	0.032	56%
flow	0.59	0.010	90%	0.62	0.012	95%	0.85	0.005	74%

The three terms above the rule are the flat ones and their q should not be read beside the rest: on BPMN only 3% of diagrams offer a crossing-reducing move and 7% an overlap-reducing one, so "crossings hold every box" is a statement about the radius and the corpus's near-trees, not about the author. The five below are live everywhere, and there the 0.00 of length and stress is a real reading: every diagram has improving moves, the mean best one worth 1.2 to 1.3 per cent of the length term and 0.5 to 0.7 per cent of stress, and none of them was taken. One such move is below perception, so at any perceptual ϵ the layouts are ϵ-stationary under the distance terms; the failure is cumulative, not local, since from the same starts the capped climber removes 0.28 to 0.31 of $C+O+L$. What a coordinate fit needs is exact stationarity, and the layouts sit a quarter of the energy from it.

The BPMN figure is the one number in the paper that the corpus's selection moves. The 300 models are the first by id of the 6,723 in the band, and on all 6,723 the alignment hold is 0.87 rather than 0.91; resampling 300 of them 2,000 times puts the committed corpus at the 99th percentile, outside the interval [0.83, 0.90] that a random 300 gives. Every other term is where it was: crossings, overlap, node–edge and $C+O$ at 1.00, length, stress and $C+O+L$ at 0.00, orthogonality 0.733 against 0.726, A3 0.312 against 0.310 and flow 0.842 against 0.854. The tables here stay on the 300 because that is the corpus the tool controls were generated for; the alignment claim is stated at 0.87, the population value. The gridiness value puts 58%, 36% and 82% of hand-placed boxes in an alignment of three or more, none of `neato`'s, 88%, 82% and 78% of `dot`'s and 80%, 69% and 75% of ELK's. The gridiness q of 0.77 to 0.78 on `neato`'s layouts, at value 1.000, is the step term's flatness, which is why alignment is compared on the value. Node–edge separation is nearly satisfied in every layout (0.000 to 0.021) and holds 0.74 to 1.00

for the same reason. Crossings hold every box at this radius; the median value on the routes is 0.000 by hand in every corpus and 0.000 for `neato`, 0.029, 0.056 and 0.000 for `dot`, 0.036, 0.066 and 0.000 for ELK. Under flow `dot` and ELK are held at 1.00 with the term at 0.000 in every corpus; the hand layouts hold 0.59, 0.62 and 0.85 at 0.131, 0.096 and 0.045, and `neato`, which knows no direction, 0.44 to 0.62 at 0.121 to 0.222.

Radius. q_d at $d \in \{0.005, 0.01, 0.02, 0.05\}$, hand layouts, then `neato`'s (`profile.py -sweep`):

	WikiPathways				Reactome				BPMN			
energy	.005	.01	.02	.05	.005	.01	.02	.05	.005	.01	.02	.05
overlap, hand	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
length, hand	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
stress, hand	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
alignment A1, hand	0.52	0.54	0.52	0.44	0.12	0.17	0.21	0.19	0.90	0.91	0.91	0.87
gridiness, hand	0.94	0.91	0.87	0.75	0.82	0.75	0.65	0.57	1.00	1.00	0.95	0.88
flow, hand	0.59	0.59	0.59	0.59	0.62	0.62	0.62	0.63	0.85	0.85	0.85	0.85
overlap, `neato`	0.60	0.60	0.60	0.60	0.75	0.76	0.76	0.76	0.41	0.41	0.41	0.43
length, `neato`	0.04	0.12	0.24	0.47	0.07	0.19	0.35	0.58	0.08	0.22	0.42	0.72
stress, `neato`	0.90	0.96	1.00	1.00	0.96	1.00	1.00	1.00	0.94	0.97	1.00	1.00
alignment A1, `neato`	0.04	0.05	0.06	0.06	0.03	0.05	0.06	0.07	0.03	0.05	0.06	0.07
gridiness, `neato`	0.93	0.86	0.78	0.65	0.90	0.83	0.77	0.61	0.92	0.85	0.78	0.67
flow, `neato`	0.50	0.50	0.50	0.50	0.45	0.45	0.45	0.45	0.62	0.62	0.62	0.62

The hand verdicts do not move with the radius. `neato`'s under length do: a smooth term near a minimum with gradient g and curvature λ holds a node when $|g| < d\lambda/2$, so a layout that is nearly a minimum is held at a large radius and free at a small one, and `neato`'s layouts, which minimise stress and not length uniformity, read as nearly uniform at $d = 0.05$ (0.47 to 0.70) and not at $d = 0.005$ (0.04 to 0.08). A hand layout at 0.00 across the range is far from the minimum at every scale the test resolves. Gridiness falls with d in every layout because a larger move reaches an alignment from further away; flow, a projection, does not move with d at all. ELK's block (`results/sweep.md`) reads as `dot`'s: length and stress 0.00 at every radius, flow 1.00 at every radius.

Reference length. Under the median edge length instead of the fitted L, `neato`'s stress layouts score 0.07, 0.08 and 0.12 under stress and the hand layouts 0.00; under length the hand layouts score 0.03, 0.04 and 0.05 and `neato`'s 0.26, 0.41 and 0.45. Under $L = 1/\sqrt{N}$ no box of any layout is held under either term, the scale being wrong for every layout. The hand verdicts do not depend on the choice; the control's does, which is why the fit is primary (Section 2).

Descent and chance. The secondary estimand, per corpus and energy: the median fraction of the cap the capped climber uses and of the energy it removes, the same for uniform sampling in the cap at the same budget and seeds, the share of diagrams where the climber ends lower on at least 12 of 15 seeds, and $q_{0.02}$ where the uncapped climber stops, the converged reference.

corpus	energy	cap used climb	random	energy removed climb	random	≥12/15	q conv.
WikiPathways	C+O+L	0.99	0.76	0.28	0.16	0.88	1.00
	C+O+A1	0.39	0.74	0.71	0.17	0.75	1.00
	stress	1.00	0.77	0.12	0.05	1.00	1.00
	C+O+F	0.44	0.76	0.27	0.18	0.77	1.00
Reactome	C+O+L	1.00	0.76	0.31	0.16	0.93	1.00
	C+O+A1	0.56	0.74	0.99	0.37	0.80	1.00
	stress	1.00	0.77	0.13	0.06	1.00	1.00
	C+O+F	0.36	0.75	0.32	0.22	0.71	1.00
BPMN	C+O+L	1.00	0.77	0.26	0.12	0.99	1.00
	C+O+A1	0.11	0.73	0.24	0.00	0.95	1.00
	stress	1.00	0.77	0.14	0.07	1.00	1.00
	C+O+F	0.16	0.74	0.09	0.04	0.66	1.00

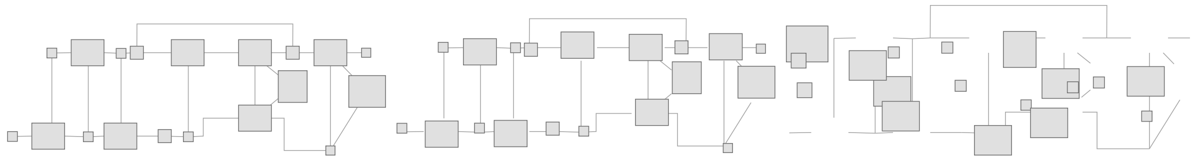
Under the distance energies the climber walks every node to the cap circle (0.99 to 1.00 of d) and is still removing energy when the budget ends; under C+O+A1 it moves 0.11 to 0.56 of the cap, the misalignments being nearer than d. The climber beats the control on all 15 seeds, which survives Holm over the corpus (exact sign $p = 3.1 \times 10^{-5}$ against 0.05/305), on 0.81 to 0.99 of diagrams under C+O+L and on all of them under stress. The converged reference's median q is 1.00 for every energy and corpus, the test's power on the sums; the fraction of diagrams whose reference stops below 0.9 is at most 0.02 everywhere but C+O+F (0.05, 0.07 and 0.13), the flow term's plateaus.

Reweighting. The linear program is degenerate as Section 2 says it must be: on the two biological corpora it puts all weight on overlap (residual per node 0.0001 and 0.0000; held-out q 0.987 [0.982, 0.993] and 0.994 [0.990, 0.998]), on BPMN all weight on crossings (0.0009; 0.992 [0.979, 0.998]), offered all eight terms on BPMN it spreads weight over six, flow 0.32, stress 0.25, orthogonality 0.22, with a held-out q of 0.801 swinging from 0.50 to 1.00 across splits: many weightings reach a high q and none is identified. Confined to length and stress it puts everything on stress and holds 0.003 to 0.011 of nodes: no distance-only weighting holds a hand layout. Confined to the four terms no layout satisfies exactly (orthogonality, alignment, node–edge, flow) it splits weight between flow and orthogonality and holds 0.22 to 0.55, except Reactome, where it rides the node–edge term and its held-out q swings from 0.10 to 0.90 across splits. The informative output is the sweep, the fraction of the corpus's nodes held as a weight w on one term joins crossings and overlap at equal weight:

term added to C+O	WikiPathways w=0.001	w=1	Reactome w=0.001	w=1	BPMN w=0.001	w=1
none ($w = 0$)	0.94		0.97		0.98	
length	0.008	0.004	0.010	0.007	0.012	0.009
stress	0.005	0.003	0.008	0.006	0.014	0.011
alignment A1	0.286	0.229	0.129	0.114	0.382	0.334
flow	0.576	0.606	0.613	0.625	0.797	0.810

The fall is a step, not a slope: whatever weight length or stress carries, 0.001 or 1, the sum holds under 0.015 of the nodes, because at almost every node some direction lowers the distance term and nothing else moves. Alignment and flow can carry weight: the criteria people partly satisfy survive in the sum at the fraction they are satisfied.

Routes. The tables above are on the stored routes; the chords are the sensitivity, `chords.md` beside the tables in the repository. Closing every edge at the box centres moves no verdict: crossings still hold every box (the WikiPathways value rises from 0.000 to 0.038, one crossing per



hand, $q = 0.90$ | displaced past $d/2$, $q = 0.05$ | snapped random, $q = 0.05$

Figure 3: The alignment controls on one BPMN model (19 boxes; the corpus medians are 0.91, 0.20 and 0.07). Left, the drawing as its author left it. Centre, every box displaced by up to 15 units, just past the $d/2$ below which the directional test cannot resolve a misalignment. Right, every box placed uniformly at random in the same bounding box and snapped to the same grid: the grid is available and the author is not. Box sizes are unchanged throughout; edges are the routes the file stores.

26 edges, the anchors' work), orthogonality falls from 0.19, 0.17 and 0.73 to 0.16, 0.05 and 0.40, and node–edge separation from 0.88, 1.00 and 1.00 to 0.76, 0.85 and 0.87. The drawn geometry only raises the hand layout's standing on the criteria it touches.

Lanes. 82% of BPMN boxes sit in a lane or pool, and a horizontal lane constrains y, so the A1 hold of 0.91 could be the container's (`results/lanes.py`). Per held box, the alignment that holds it: 30% a column, which no horizontal lane imposes; 13% a row shared with a box of another lane or of no lane, which no lane imposes either; 57% a row within one lane. Those lanes are a median 6.3 box heights tall (quartiles 3.7 and 9.4; 3% under two box heights), so the row placement inside them was chosen, not forced. At most the choice of row count is the lane's; where the box sits on that row, and the 43% held across or against the lanes, is the person's.

The paired tests. Per corpus, hand against `dot` on each criterion alone: the Hodges–Lehmann shift of $q_{0.02}$ with its 95% interval, zeros dropped, and the Holm-adjusted signed-rank p within the corpus's family (`analyse.py`; the same families against `neato`, `prism` and ELK are in `results/tests/`).

energy	WikiPathways HL [95%]	p	Reactome HL [95%]	p	BPMN HL [95%]	p
crossings	−0.04 [−0.07, −0.01]	0.016	+0.02 [−0.01, +0.05]	0.30	+0.10 [+0.08, +0.12]	<1e-4
overlap	−0.14 [−0.18, −0.11]	<1e-4	−0.07 [−0.10, −0.06]	0.0024	−0.22 [−0.29, −0.11]	<1e-4
length	−0.01 [−0.03, +0.00]	0.27	−0.03 [−0.05, −0.01]	0.017	−0.09 [−0.12, −0.06]	<1e-4
stress	−0.03 [−0.04, −0.00]	0.022	−0.03 [−0.05, −0.01]	0.0048	−0.08 [−0.11, −0.05]	<1e-4
orthogonality	+0.01 [−0.00, +0.04]	0.27	+0.02 [+0.00, +0.04]	0.036	+0.43 [+0.41, +0.45]	<1e-4
alignment A1	−0.46 [−0.49, −0.43]	<1e-4	−0.70 [−0.73, −0.67]	<1e-4	+0.04 [−0.01, +0.09]	0.14
alignment A3	−0.25 [−0.28, −0.22]	<1e-4	−0.28 [−0.31, −0.26]	<1e-4	+0.13 [+0.10, +0.16]	<1e-4
gridiness	−0.13 [−0.15, −0.11]	<1e-4	−0.29 [−0.32, −0.26]	<1e-4	+0.21 [+0.17, +0.25]	<1e-4
node–edge	+0.13 [+0.10, +0.16]	<1e-4	+0.18 [+0.15, +0.21]	<1e-4	+0.26 [+0.23, +0.28]	<1e-4
flow	−0.40 [−0.42, −0.38]	<1e-4	−0.33 [−0.35, −0.31]	<1e-4	−0.17 [−0.19, −0.14]	<1e-4

The shifts under crossings, overlap, length and stress are distinguishable but confined to the minority of diagrams where the pair differs at all. The large shifts are the profile's gaps: the hand layout sits below `dot` on alignment in the biological corpora (−0.46 and −0.70) and level with it on BPMN ($p = 0.14$), below it on flow everywhere (−0.17 to −0.40), and above it on node–edge separation and, in BPMN, on orthogonality (+0.43), the criteria `dot` trades away.

The fourth corpus, restored. The pre-registration named HOLA's formative drawings [13] and the previous draft dropped them without a word. Restored: 136 drawings, 17 participants each laying out the same 8 abstract graphs, scraped from the study's published SVGs (`data/hola.txt`, provenance beside it), chord edges, below this paper's 15-box band and free of every diagram notation. The pattern holds: overlap and crossings 1.00, length and stress 0.00, alignment 0.50,

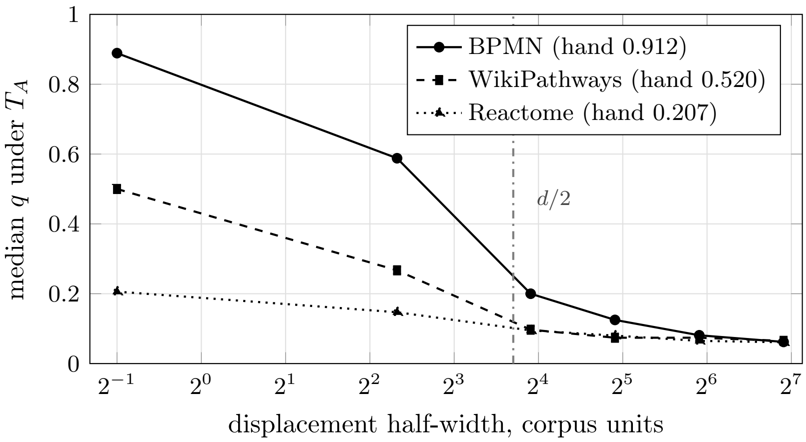


Figure 4: What the jitter sweep sees. The declared half-pitch test sits at the left edge, 0.5 units, where nothing has changed. The corner of T_A protects a single displaced box below $\epsilon = d/2$, but every box is displaced, relative offsets reach 2ϵ, and the hold falls well before $d/2$: BPMN 0.89 to 0.59 by a half-width of 5. By $d/2$ every corpus is at the grid-snapped level of 0.06 to 0.08.

between Reactome's 0.21 and BPMN's 0.91. Node–edge holds 1.00, so H4 fails here too. These are drawings from a study tool rather than a diagram editor, which bears on the guide question of Section 7, though whether that tool drew alignment aids is not recorded in the corpus.

Null layouts. Review asked for the chance line under every term, not only alignment. Uniform random layouts of the same graphs, chord geometry, one draw per diagram, seeds from the diagram ids: alignment 0.06 to 0.07 against hand 0.52, 0.21 and 0.91; node–edge 0.11 to 0.12 against 0.76 to 0.87; flow 0.26 to 0.41 against 0.59 to 0.85; orthogonality 0.00 against 0.05 to 0.40. Length and stress hold 0.00 in the random layouts too: no layout of any provenance is stationary under a distance term at this radius except a tool's converged output, so the distance result separates optimizers from everyone, not humans from tools. Crossings and overlap hold 0.58 to 0.67 of random boxes, the flat base rate the active column already prices. Under a jitter of one grid cell the alignment hold keeps 0.76, 0.44 and 0.19 of its three corpora while orthogonality falls from 0.40 to 0.18 and 0.16 to 0.06: orthogonality, not alignment, is the tie artifact. Axis-only moves, which preserve every perpendicular alignment tie, still improve length and stress on the median diagram of every corpus (q 0.00, means 0.005 to 0.016), and a weight of 0.001 on length collapses the held fraction of $C{+}O{+}A_1$ from 0.89, 0.51 and 0.22 to 0.13, 0.11 and 0.07: the distance terms are unpriced even inside the alignment-preserving move set, and no small weight on them survives beside alignment. The generator and every number: `results/nulls.py`.

The pre-registration's scorecard. H1, crossings and overlap hold the hand layout above 0.9 and above `neato`: holds, 1.00 against 0.41 to 0.76. H2, length alone holds under 0.2 and below `neato`: holds, 0.00 against 0.24 to 0.42. H3, no positive length weight repairs $C{+}O{+}L$: holds, fitted weight 0.000 everywhere and q 0.008 at any weight past 0.001. H4, alignment holds more boxes than orthogonality or node–edge in every corpus: **refuted**. Node–edge holds 0.88, 1.00 and 1.00 of hand-placed boxes against alignment's 0.52, 0.21 and 0.91. Node–edge is priced in the standard energy and alignment is not, so the omitted-term conclusion stands, but the registered ranking was wrong and the recorded prediction with it. H5, exploratory, stress holds the hand layout less than `neato`'s: holds, 0.00 against 1.00. The registered fourth corpus, HOLA's formative drawings, was dropped undeclared from the previous draft; it is restored above. Registered and still not run: the deduplication pass, the 41–100 band, the caps in pitch units.

7 What else could produce this

1. **The editor snapped to a grid**, so alignment is the tool's. Excluded, by the control the pre-registration named and by a stronger one it did not. The detected pitch is one unit in Reactome and BPMN (67% and 63% integer coordinates) and none in WikiPathways (17%, against 0.37 of boxes exactly aligned in x). Displacing every box by up to half a pitch leaves the held fraction at 0.500, 0.206 and 0.889 against 0.520, 0.207 and 0.912. That test cannot fail at this pitch, and the reason is the instrument: T_A is $\min_j \min(|\Delta x|, |\Delta y|)$, so a box offset by ϵ from its nearest agreement is released only when a move of radius d lowers the term, and moving d toward the agreement lands at $d - \epsilon$, which is worse whenever $\epsilon < d/2$. Half a pitch is 0.5 units against a $d/2$ of 8, 8 and 13. The control that does discriminate is a layout snapped to the same grid, with the same boxes and the same bounding box and no author: it holds 0.065, 0.075 and 0.067. Displacing the hand layouts past $d/2$ takes them to that same level (0.097, 0.095 and 0.200 at a half-width of 15, 0.065, 0.061 and 0.062 at 120). The grid is available to the random layout and does not produce the effect. Two additions after review: a random layout snapped to the plausible working pitch of five units holds 0.076, the same 0.07 as at pitch one, so the null is not flattered by the finest pitch; and the jittered hand layouts begin falling well below $d/2$ (0.59 on BPMN at a half-width of 5), so the corner argument bounds only the declared half-pitch test, and the caption of Figure 4 now says so.

2. **The editor snapped to the neighbours**, so alignment is the guide's. Not separable on these corpora, and conceded. A grid control cannot see snap-to-object guides, which PathVisio and the BPMN editors draw on every drag and Reactome's curation tool does not, and the alignment hold orders exactly as those affordances do: 0.87 to 0.91 where guides meet every drag, 0.52 where they are offered, 0.21 where they are absent. For fitting an energy the attribution changes nothing, the coordinates hold a term the energy lacks either way; for the word "optimize" it does, and the claim here is about the layouts, not the muscle. Separating them needs a guide-free editor or an interaction log, and these corpora carry neither. The nearest evidence is the restored HOLA corpus: drawings from a study tool, not a commercial editor, hold alignment at 0.50, so the phenomenon does not need Signavio to appear.

The rest, each with the test that excluded it and the numbers it turned on:

alternative	test	result
Reference length	L at the median edge length and at $1/\sqrt{N}$	hand held on 0.00 to 0.06 of boxes under length and stress, as under the fit; only the control moves
Straight segments where the person drew polylines	primary tables on the stored routes, chords beside them	no verdict moves
A constrained energy whose constraints are absent	the flow term in the tables; the BPMN lanes decomposed	flow holds 0.59 to 0.85 of hand-placed boxes, so reading direction is a criterion partly satisfied and not the missing constraint; 43% of held BPMN boxes are held by an alignment no horizontal lane can impose, and the lanes of the rest leave a median 6.3 box heights of room. Containment as a hard constraint was not added
The radius	$d \in \{0.005, 0.01, 0.02, 0.05\}$; at $d = 0.02$ the move is about one grid step	the hand verdicts are the same at every d. The table at d in units of box width and of median edge length was not run
Selection: near-trees make overlap and crossings easy	q against m/n over diagrams (`sparsity.py`)	crossings do track density (Spearman ρ -0.37, -0.34, -0.11 by hand and -0.48, -0.30, -0.41 for `dot`), these being near-trees at m/n 1.06, 1.04 and 1.11 against 1.30 in the Rome benchmark [21], so "crossings hold every box" is partly the trees' gift, given to every layout of the same graphs alike; crossings are what the theory is about [1], and on these diagrams there are few to remove. Overlap does not (+0.02, -0.06, +0.12), and length and stress are 0.00 in the sparse and the dense half both. No hand–tool difference is explained
The tools are the wrong tools	ELK layered 0.12.0, the engine the BPMN and KIELER editors run, as a fourth control	its profile is `dot`'s: length and stress 0.00, flow 1.00 at value 0.000, alignment 0.42 to 0.62 where `dot` reaches 0.89 to 1.00

8 Related work

Aesthetic criteria and their weighting go back to Tamassia, Di Battista and Batini [3] and were made an explicit energy by Davidson and Harel [2]. Purchase [4] gave the criteria their metrics, and that list has no alignment metric. Whether people agree with the criteria has been asked three ways.

work	what it did	what it gives, and what it leaves
By having people draw		
van Ham, Rogowitz [16]	hand layouts against a force-directed algorithm	hand layouts carry fewer crossings
Dwyer et al. [17]	32 participants laid out one graph; 194 judges ranked the results against three automatic layouts	judges' choices track stress, over twelve layouts and without a test
Purchase, Pilcher, Plimmer [25]	participants drew from adjacency lists	crossing removal the strongest aesthetic, alignment to a grid important
Kieffer et al. [13]	17 designers' layouts of 8 small graphs, ranked by 66 judges	gridiness correlates with rank on all 8, which is where HOLA's alignment step comes from
Mooney et al. [11]	preference and shortest-path performance against stress, graphs to 50 nodes	people prefer low stress and perform better as it falls; why the conclusion adds a term and removes none
Mooney et al. [12]	4,890 drawings harvested automatically from 27 editions of these proceedings	the nearest corpus, not used here: a figure in a graph-drawing paper is as likely an algorithm's output, provenance is not per drawing, and edge directions are dropped
By showing people drawings		
Purchase et al. [18]	a Turing test on hand and algorithmic drawings	hand drawings distinguishable, and judged better
Klammler, Mchedlidze, Pak [14]	Nelder–Mead fit of a linear aesthetic combination discriminating good from worsened layouts	no human coordinates enter it; they call human-labelled data a critically important accomplishment still to come
By learning from examples		
Masui [15]	a weighted constraint combination from good and bad layouts, by genetic programming	neither tested whether the examples are minima of the learned energy
Spönemann et al. [26]	weights adjusted from user selections among candidates	as above
Flud [19]	annealing continued from crowd layouts of pathway diagrams, three networks	the closest prior descent from a hand layout; it raised an asserted five-term score and read that as success. The cap, the per-term test, the reference and the controls here are what make it a measurement of the score
Zhang et al. [28]	64,222 pairwise preference labels	the nearest recent work; it learns from preferences, not coordinates
O'Donovan, Agarwala, Hertzmann [8]	a 122-parameter energy over single-page graphic designs fitted to a handful of human examples by non-linear inverse optimisation	each example taken as optimal under the unknown parameters: the assumption examined here, and the closest prior instance of it we found

Alignment as a human criterion is therefore known since 2012. What is new here is the count of how many hand-placed boxes it holds, on diagrams of 15 to 40 boxes, beside the counts for the criteria tools weight. Two results bound what such a count can mean: van Wageningen, Mchedlidze and Telea [20] show drawings with identical metric values that differ in quality, so a fitted energy with no residual would still not be the energy people use, which is why the claims here stay on the negative side; and Kobourov, Pupyrev and Saket [27] found crossings matter little on sparse large graphs, consistent with the sparsity result.

The machinery is standard elsewhere. Inverse optimisation, the recovery of an objective from solutions taken as optimal, is due to Ahuja and Orlin [22] for linear programs and to Keshavarz, Wang and Boyd [23] for the convex case; Chan, Lee and Terekhov [24] define the goodness-of-fit residual reported here. Matched-budget random controls are standard in hyperparameter search [6] and were prescribed as the baseline for genetic algorithms in 1995 [7]; we have not found them in layout. No paper fitting energy weights to human-placed coordinates was found in four searches of 2024 to 2026 work.

9 Reproducibility

Parsers, the energy (one function per term), the directional test, the descent, the fitting scripts, the parsed corpora and every table's script are in the repository under `example/diagrams`. The directional test is deterministic; the descents are deterministic from their seed on every platform. Every table is from the battery of 2026-08-23 on commit `1987633`, run by `results/run_all.sh` followed by `results/tests.sh`, `results/lanes.py` and `results/sparsity.py`; `results/manifest.txt` records the platform, the tool versions (gcc 13.3.0, Graphviz 2.43.0, elkjs 0.12.0, HiGHS via scipy 1.18.1), the seeds and the SHA-256 of the three raw archives and of the eighteen corpus files as measured. The population number of Section 6 reads the tracked `data/bpmn_population_a1.csv`, 6,723 rows regenerated from the archive by the commands in its header, so the abstract's 0.87 rebuilds from a clone. The BPMN archive is distributed for academic use; the repository carries only derived coordinates and ids, never model content, and the two biological corpora are CC-licensed exports.

10 Conclusion

Coordinates a person accepted are not minima of the energy in use, and the way they fail is specific: exact under overlap, nowhere near under uniform edge length or stress, and held instead by an alignment into rows and columns that, HOLA's gridiness step aside [13], no standard energy prices. On every criterion measured, a hand layout reads as a layered layout with partial alignment and partial flow.

A method that learns weights from such layouts must offer alignment among its terms or it fits nothing the person did. For a tool builder that is one addition and one number, not a subtraction: edge length and stress keep their weight, since a term inert at human coordinates is not thereby unwanted and the preference evidence runs the other way [11, 17]. Coordinates and preferences are different evidence about the same people, and only the first is measured here. The number to report is the fraction of boxes the energy holds when started from a layout a person accepted.

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